

High Frequency Transformer (HFT) Design and Simulation in ANSYS Maxwell

- EE65 (N87 Ferrite) Transformer for 5 kW DAB Converter

Objective:

- Design a **5 kW High Frequency Transformer**
- Operate at **100 kHz switching frequency**
- Validate transformer design using **ANSYS Maxwell transient simulation**
- Verify:
 - Flux density limit (0.2 T)
 - Induced voltages
 - Open circuit behavior

Transformer Design Parameters:

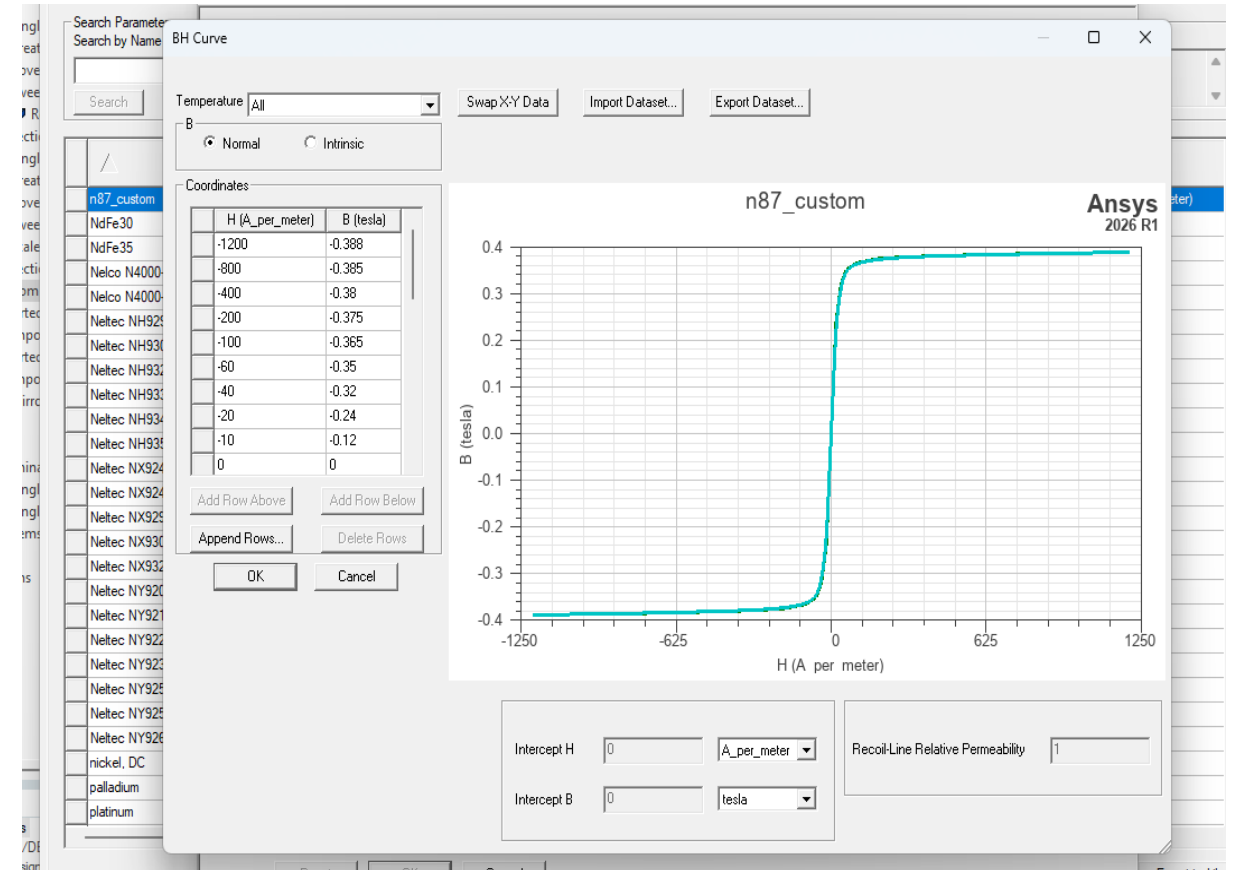
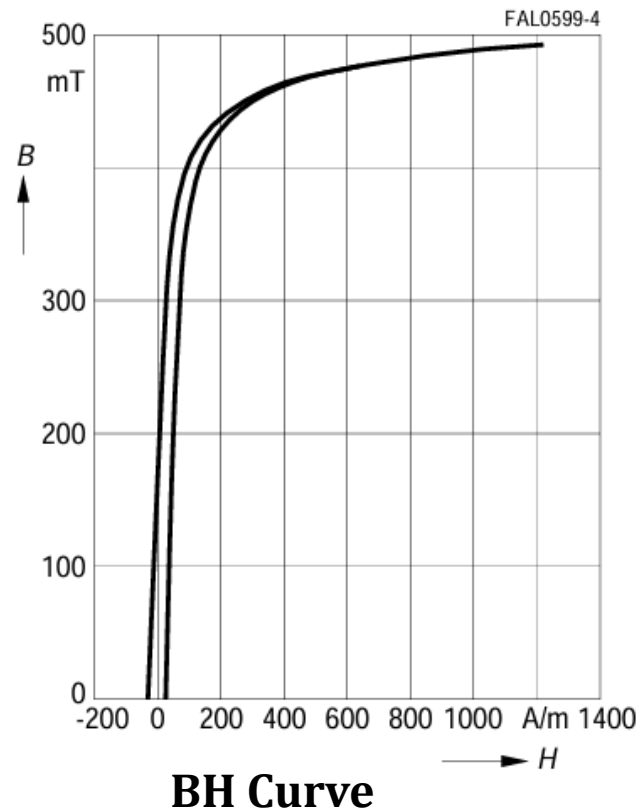


A 5kw transformer was designed for a max flux density =0.2T, at 100kHz.
EE65 core was selected by Area Product method

Parameters	Value
Input Voltage Range	180 – 270 V
Nominal Input Voltage	240 V
Output Voltage Range	300 – 420 V
Nominal Output Voltage	360 V
Power rating	5kW
Switching Frequency	100 kHz
Primary Turns	6
Secondary Turns	9
Peak Flux Density	0.2T
Core	EE65(N87)
Windings	Litz(6mm ²)

N87 Ferrite Material Creation:

Since default N87 material was unavailable, a custom material model (**N87_custom**) was created using manufacturer datasheet values.

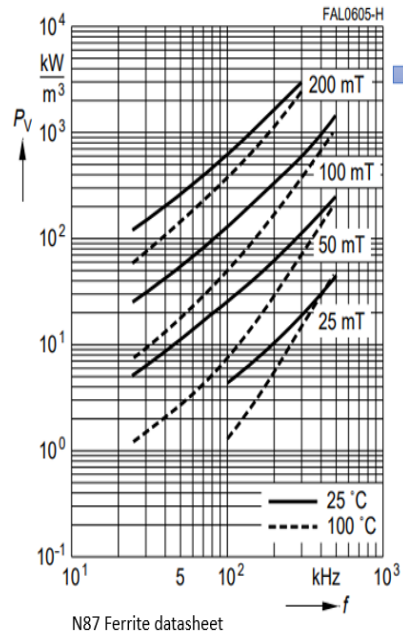


Nonlinear B-H characteristics of the N87 ferrite material were imported to accurately model magnetic saturation and hysteresis effects.

Steinmetz Core Loss Modelling:

Steinmetz parameters were extracted from the N87 datasheet at 100 kHz and implemented in Maxwell

Extracting Steinmetz Parameters from Datasheets:

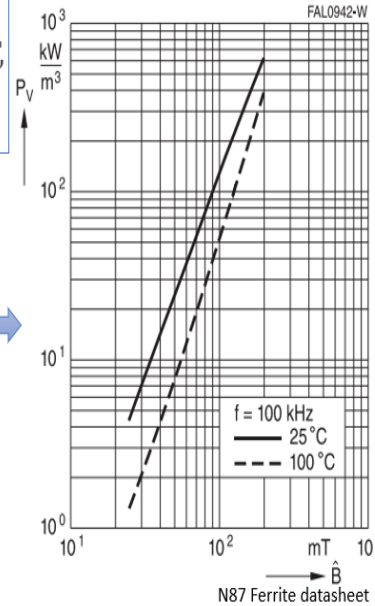


Frequency Exponent α :
Log-log slope of P_v vs f at $B = 0.2$ T,
100°C $\rightarrow \alpha \approx 1.45$

Flux Density Exponent β :
Log-log slope of P_v vs B at $f =$
kHz, 100°C $\rightarrow \beta \approx 2.8$

Steinmetz Coefficient k :

Using $P_v = 375$ kW/m³ at 100 kHz,
 $B = 0.2$ T $\rightarrow k \approx 42.774$ kW/m³
 $k_{eq} = 0.00191$ kW/m³ (in Hz)



$k \rightarrow C_m$ (f in Hz)

$\alpha \rightarrow X$

$\beta \rightarrow Y$

View / Edit Material

Material Name: n87_custom Material Coordinate System Type: Cartesian

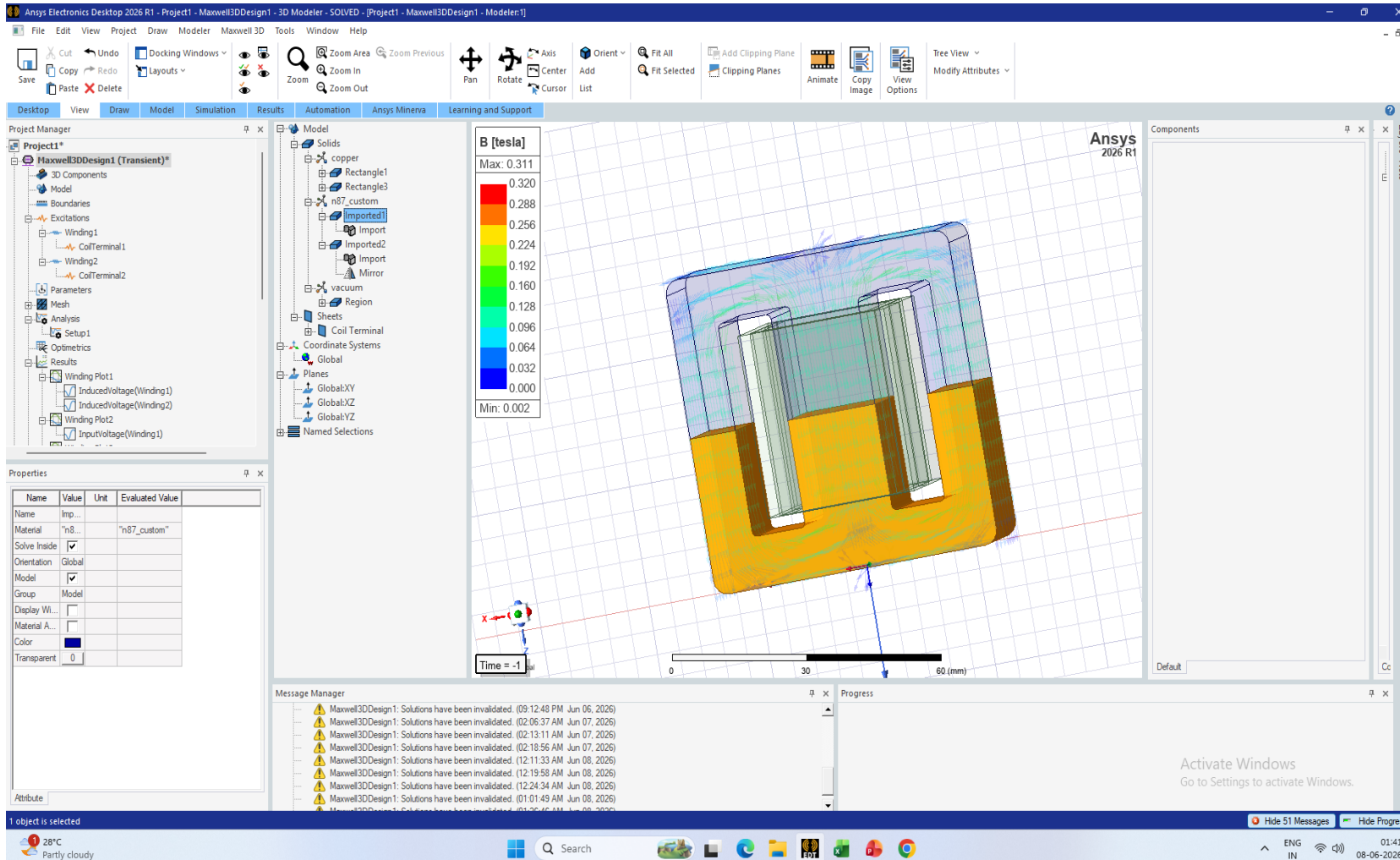
Properties of the Material

Name	Type	Value	Units
Relative Permeability	Nonlinear	B-H Curve...	
Bulk Conductivity	Simple	0.1	siemens/m
Magnetic Coercivity	Vector		
- Magnitude	Vector Mag	0	A_per_meter
- X Component	Unit Vector	1	
- Y Component	Unit Vector	0	
- Z Component	Unit Vector	0	
Core Loss Model		Power Ferrite	w/m^3
- Cm	Simple	1.91	
- X	Simple	1.45	
- Y	Simple	2.8	
- Kdc	Simple	0	
- Equiv. Cut Depth	Simple	0	meter
Mass Density	Simple	4850	kg/m^3
Composition		Solid	
Young's Modulus	Simple	0	N/m^2
Poisson's Ratio	Simple	0	
Magnetostriction	Custom	Edit...	
Inverse Magnetostriction	Custom	Edit...	

$$P_v = C_m f^X B^Y$$

Transformer Geometry (core)

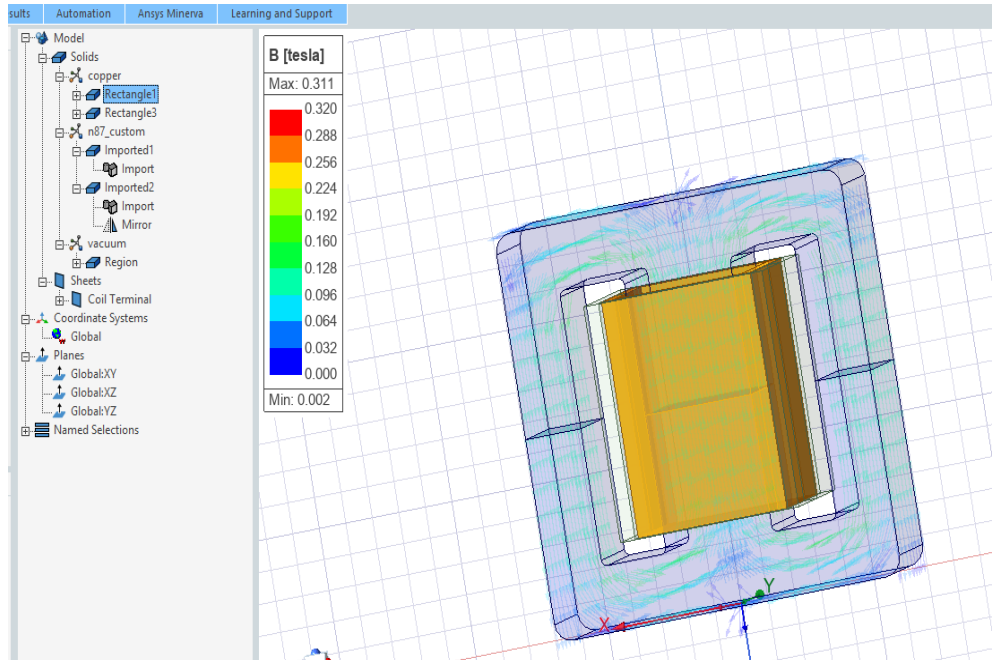
The Same Transformer was designed in Ansys Maxwell in Transient mode (For high frequency and Arbitrary Excitations).



The transformer geometry was modeled in ANSYS Maxwell using the designed dimensions.

EE65 core was designed and N87 custom core material was assigned to it

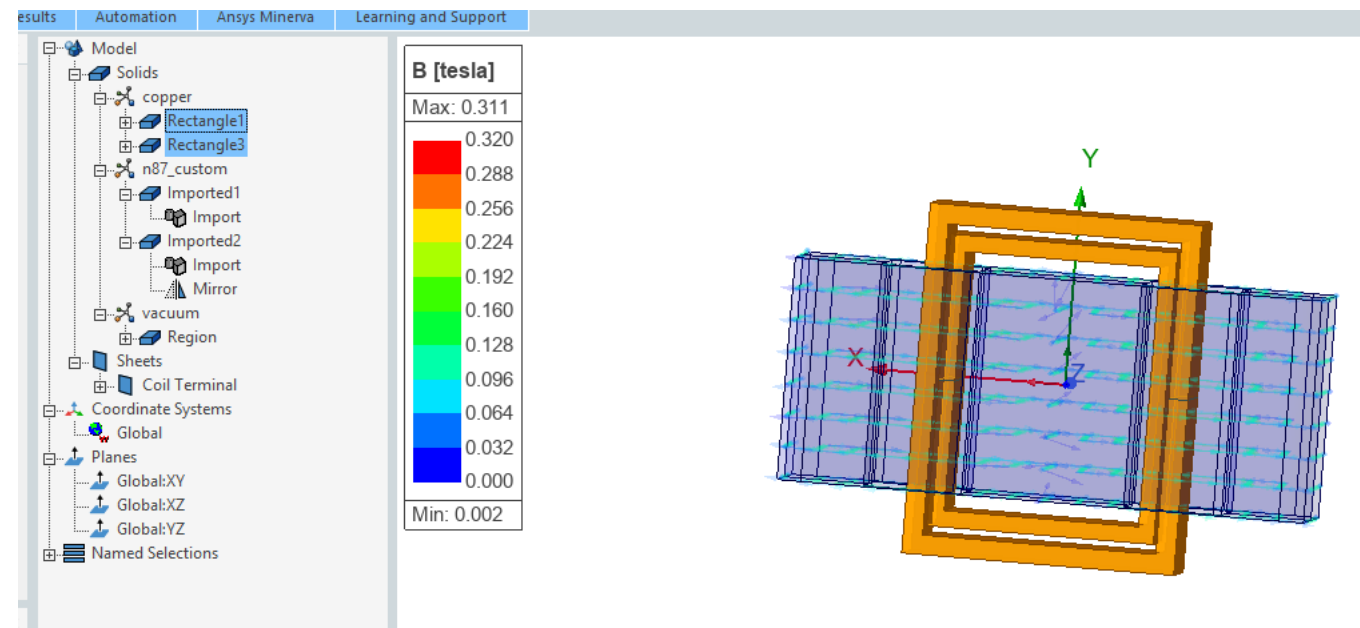
Transformer Geometry (Windings)



Front View



Rectangle 1 and rectangle 3 are the primary and secondary windings swept over middle limb of core

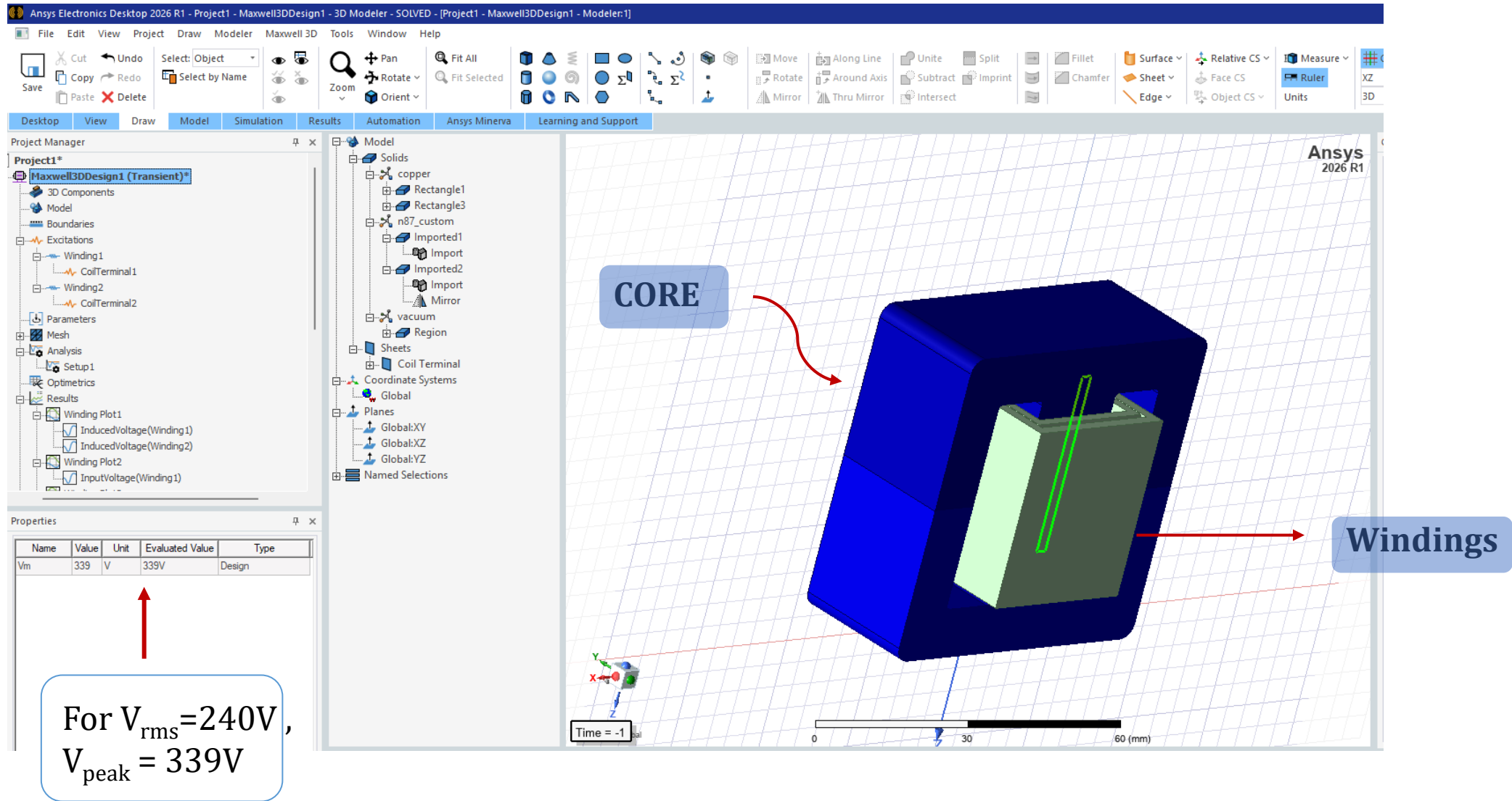


Top View



- Primary turns = 6
 - Secondary turns = 9
- Copper material was assigned to the windings

Complete Transformer Model:



Complete geometry of the Transformer (Core+ windings)

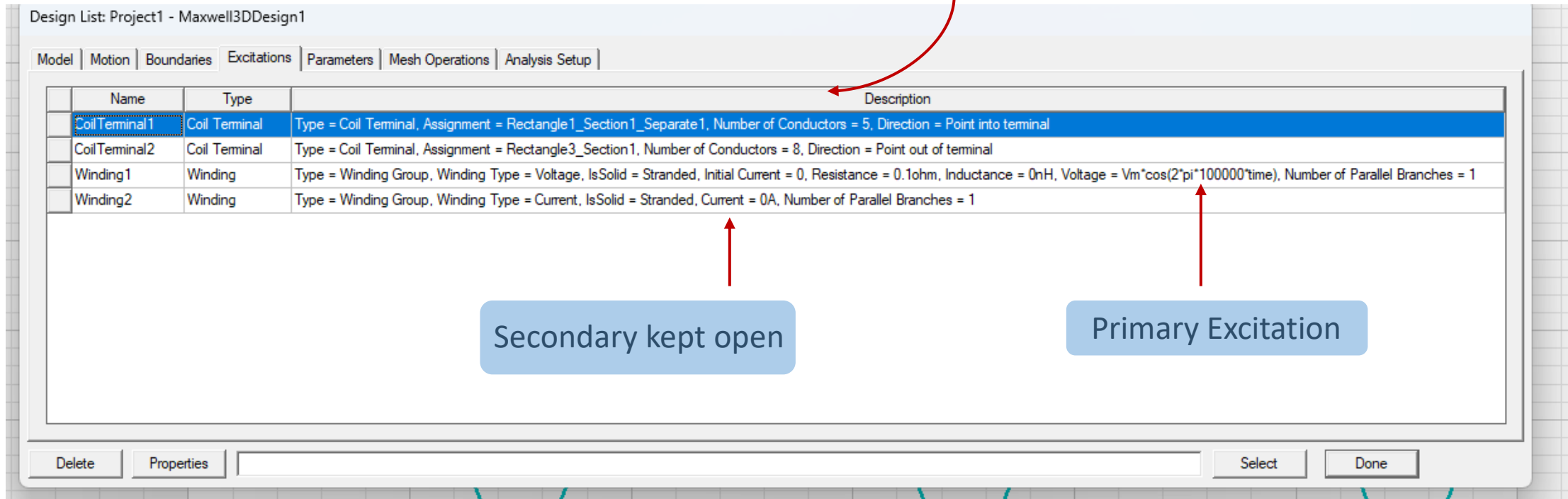
Simulation Setup:

Parameter	Value
Solver	Maxwell Transient (high frequency and arbitrary excitation)
Excitation	Voltage Source
Frequency	100 kHz
Analysis Type	Open Circuit
Secondary Condition	Open
Simulation Cycles	3
Time Step	0.2us

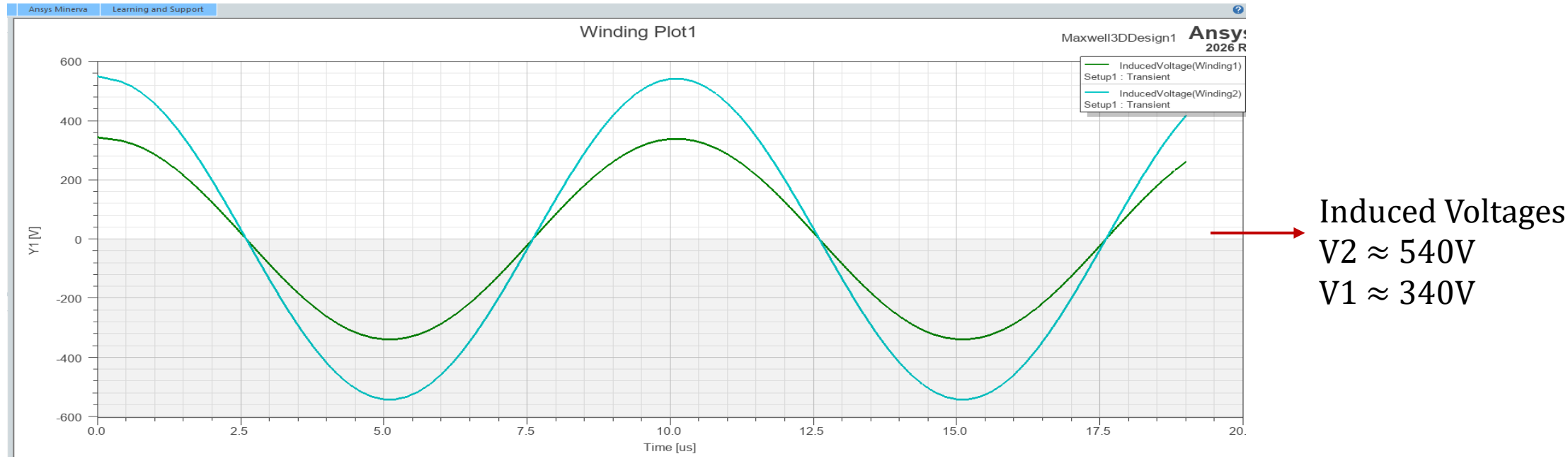
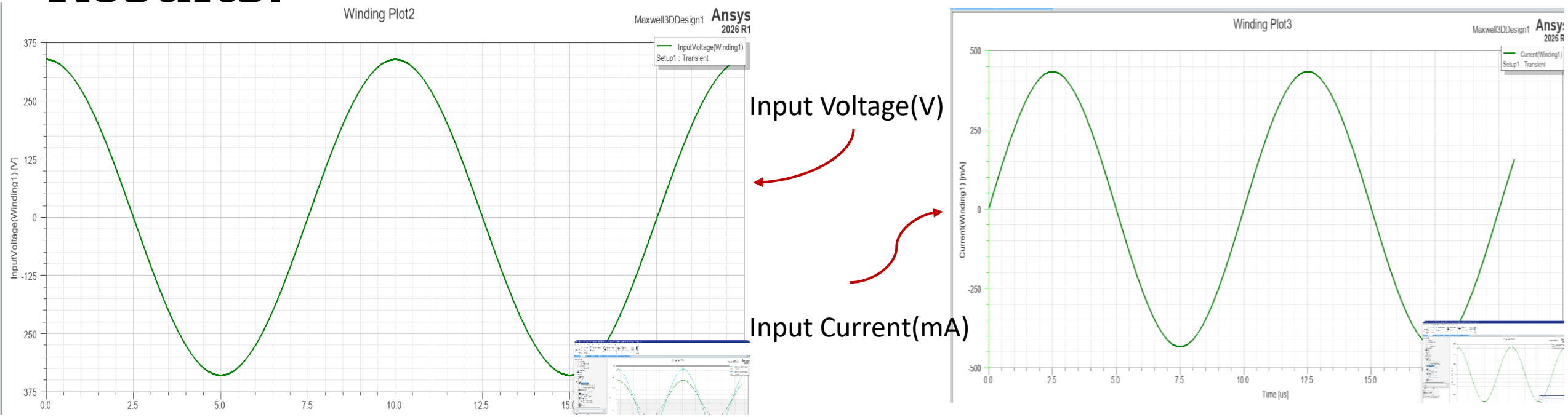
Open Circuit Test – Sinusoidal Excitation:

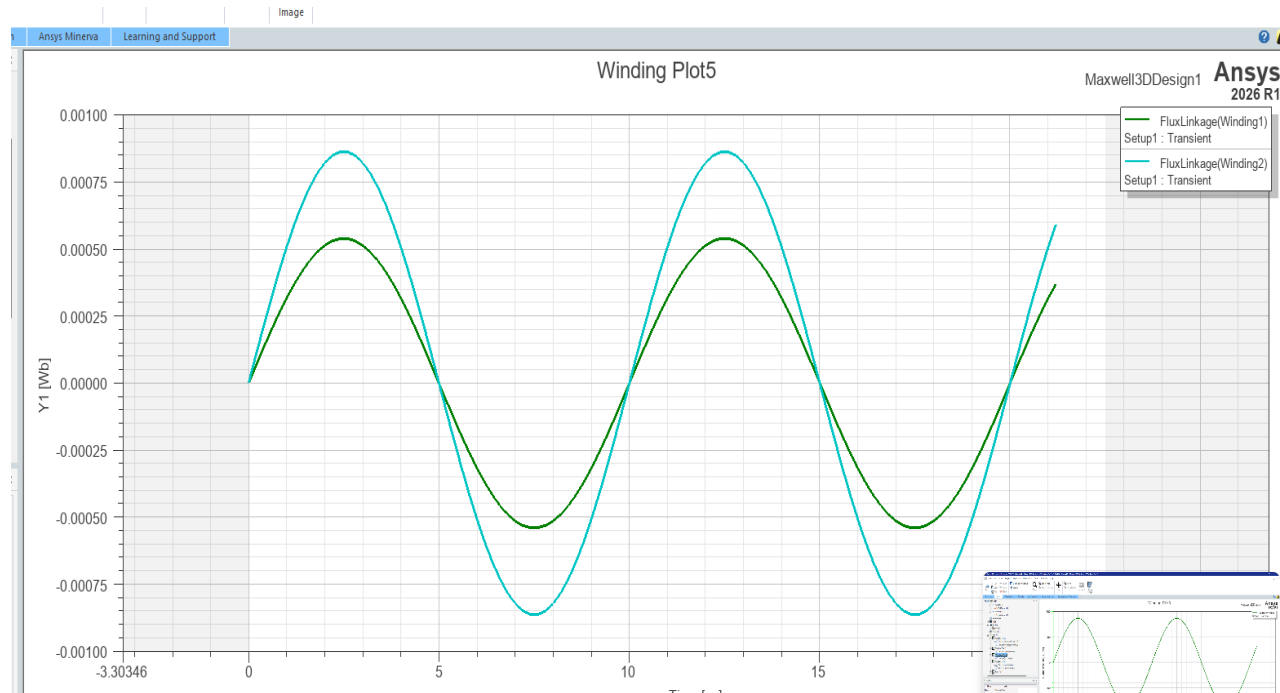
Primary excited with Sinusoid of peak V_m ($V_m = \sqrt{2} \times 240 \approx 339\text{ V}$) and secondary was kept open(0A) for **open circuit test**

$N_p=5$
 $N_s = 8$

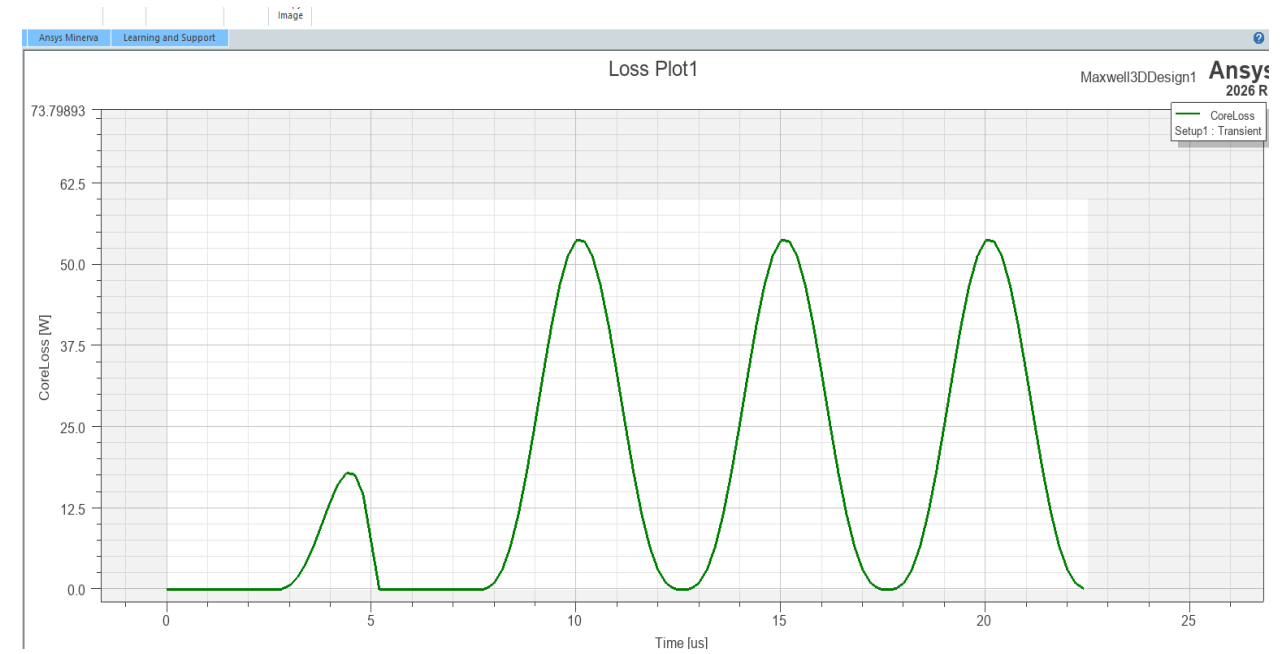


Results:



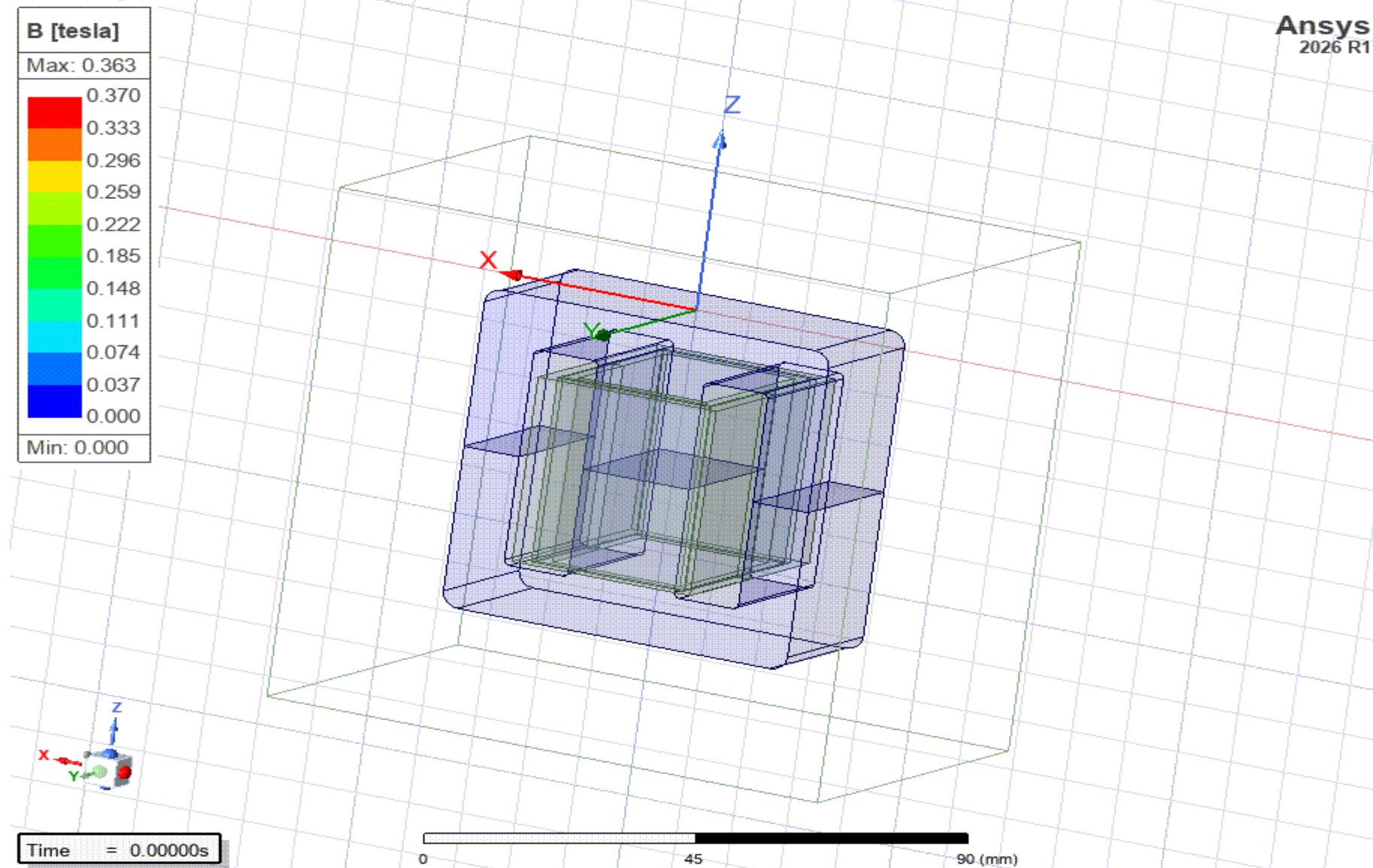


Flux Linkages



Core Loss plot

Field Distribution in the core:



Peak flux density in most regions of the core remains close to the design limit of **0.2 T**, validating the transformer design.

Observations:

- Peak flux density ≈ 0.2 T
- Expected voltage transformation observed ($V_2/V_1 \approx N_2/N_1 \approx 1.5$)
- Magnetizing current validated
- Core behavior matches design assumptions

Open Circuit Test — Square Wave Excitation:

Primary excited with Square wave of voltage **240V** and secondary was kept open(0A) for **open circuit test**

Ns = 8
Np = 5

Design List: Project1 - Maxwell3DDesign1

Model | Motion | Boundaries | Excitations | Parameters | Mesh Operations | Analysis Setup

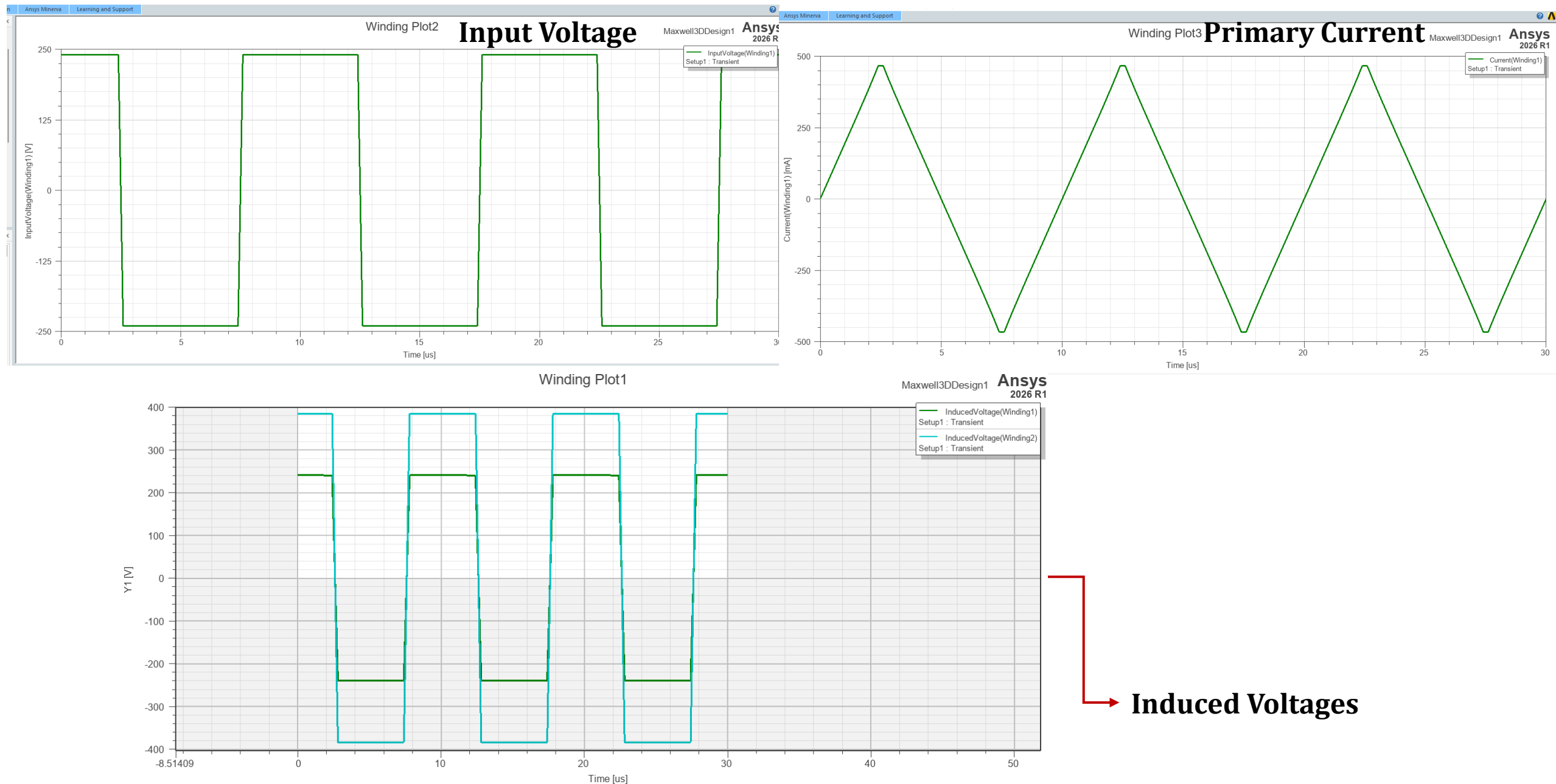
Name	Type	Description
Coil Terminal1	Coil Terminal	Type = Coil Terminal, Assignment = Rectangle1_Section1_Separate1, Number of Conductors = 5, Direction = Point into terminal
Coil Terminal2	Coil Terminal	Type = Coil Terminal, Assignment = Rectangle3_Section1, Number of Conductors = 8, Direction = Point out of terminal
Winding1	Winding	Type = Winding Group, Winding Type = Voltage, IsSolid = Stranded, Initial Current = 0, Resistance = 0.1ohm, Inductance = 0nH, Voltage = Vm*sgn(cos(2*pi*100000*time)), Number of Parallel Branches = 1
Winding2	Winding	Type = Winding Group, Winding Type = Current, IsSolid = Stranded, Current = 0A, Number of Parallel Branches = 1

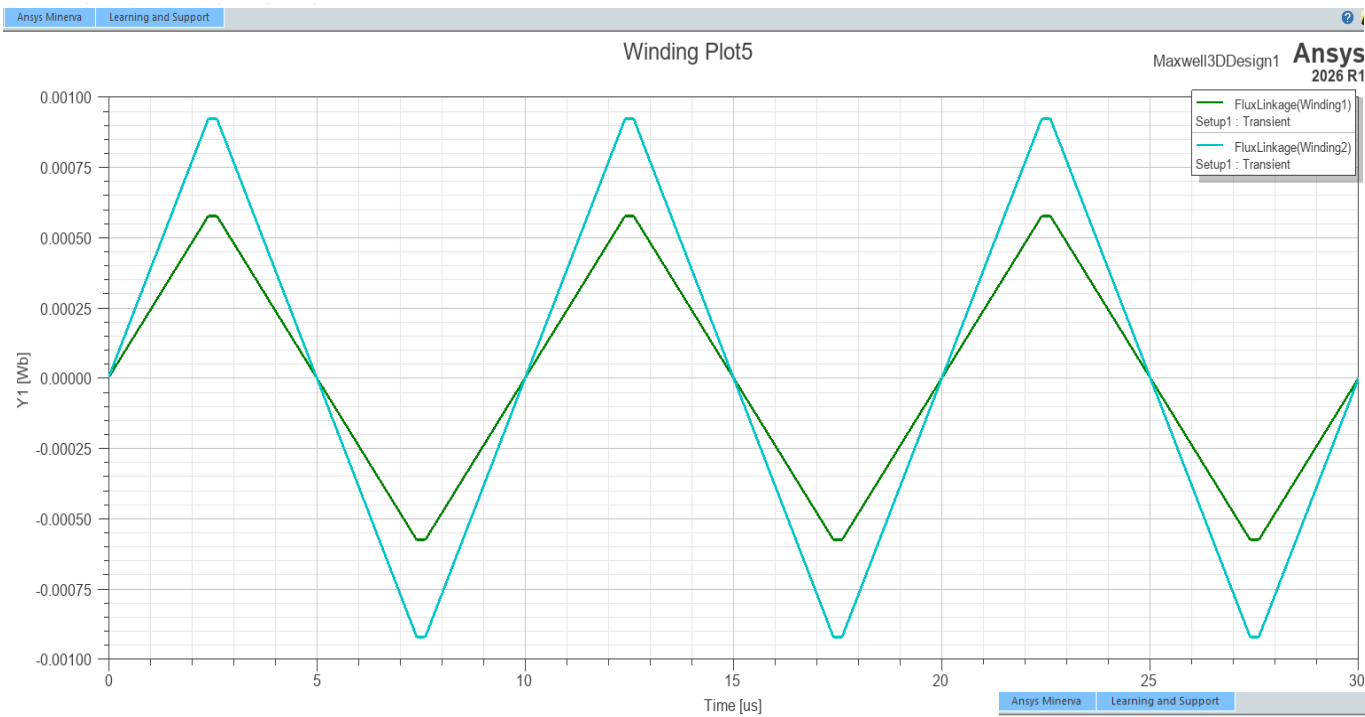
Secondary open

Primary Excitation

Delete Properties Select Done

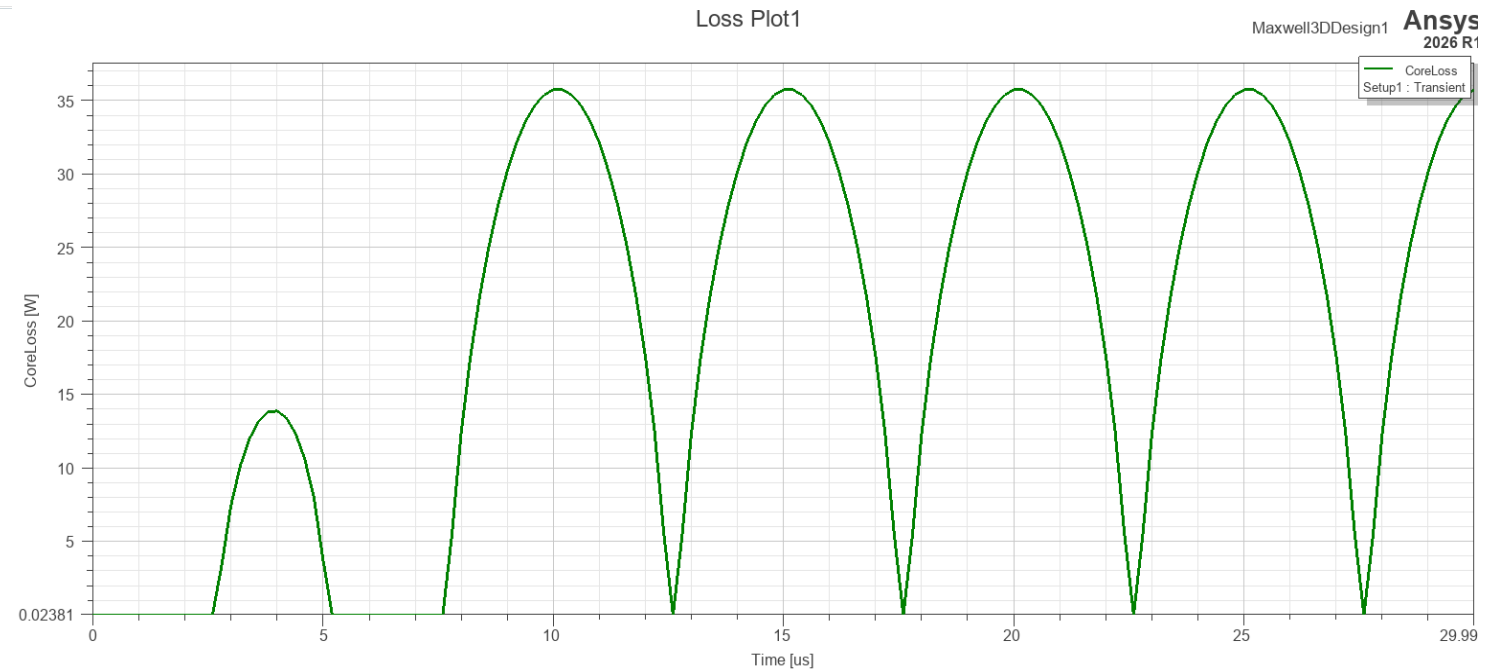
Results — Square Wave Excitation:



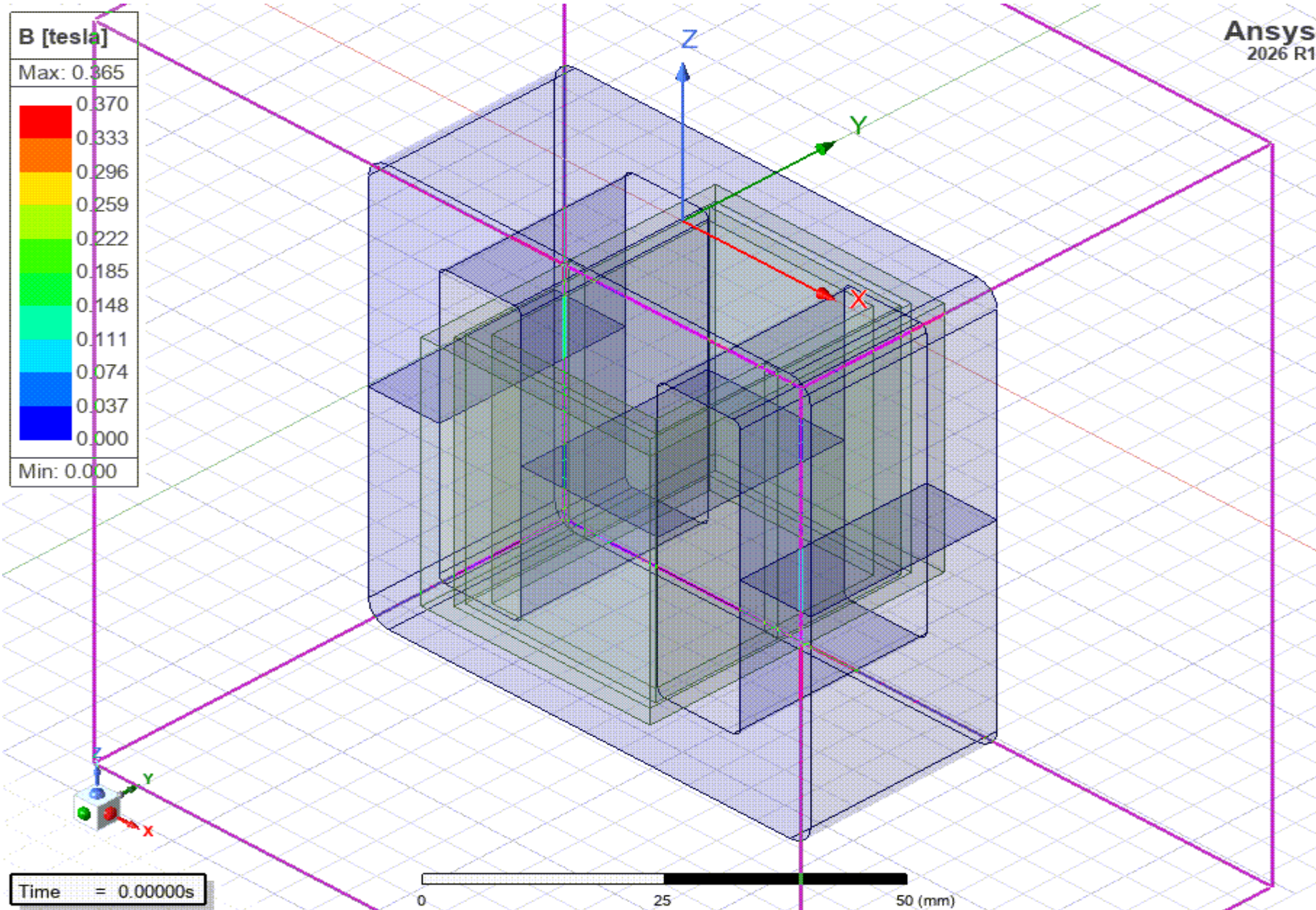


Flux Linkages

Core Loss Plot



Field Distribution in the Core:



Here also the peak flux density in the core is at almost **0.2T**, which was taken while designing the hardware.

Observations

- Transformer operates under DAB-relevant excitation
- Flux remains within saturation limit
- Voltage waveform consistency verified

Conclusion:

- A 5 kW, 100 kHz EE65 high-frequency transformer was designed.
- N87 ferrite material was modeled using custom Steinmetz parameters.
- Transient simulation performed in ANSYS Maxwell.
- Open-circuit performance validated under sine and square-wave excitation.
- Peak flux density remained near the design target of 0.2 T.
- Initial validation of transformer design successfully completed.

THANK YOU